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1 Introduction to linear transformations

1.1 Images of functions

Consider a function

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

The domain of T is $\mathbb{R}^n$ and the co-domain of T is $\mathbb{R}^m$. The case $n = m$ is allowed, of course. In engineering or physics, the domain is sometimes called the input space and the co-domain is called the output space. Using this terminology, the points $\mathbf{x}$ in the domain are called the inputs, and the points $T(\mathbf{x})$ produced by the mapping are called the outputs.

Definition 1. The vector $\mathbf{b} \in \mathbb{R}^m$ is in the range of T , or in the image of T , if there exists some $\mathbf{x} \in \mathbb{R}^n$ such that $T(\mathbf{x}) = \mathbf{b}$.

In other words, $\mathbf{b}$ is in the range of T if there is an input $\mathbf{x}$ in the domain of T that outputs $\mathbf{b} = T(\mathbf{x})$. In general, not every point in the co-domain of T is in the range of T . For example,

consider the vector mapping $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as

$$T(\mathbf{x}) = \begin{bmatrix} x_1^2 \sin(x_2) - \cos(x_1^2 - 1) \\ x_1^2 + x_2^2 + 1 \end{bmatrix}.$$

The vector $\mathbf{b} = (3, -1)$ is not in the range of T because the second component of $T(\mathbf{x})$ is positive. On the other hand, $\mathbf{b} = (-1, 2)$ is in the range of T because

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1^2 \sin(0) - \cos(1^2 - 1) \\ 1^2 + 0^2 + 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \mathbf{b}.$$

Hence, a corresponding input for this particular $\mathbf{b}$ is $\mathbf{x} = (1, 0)$.

1.2 Linear transformations

Definition 2. Let V and U be vector spaces over the scalar field $\mathbb{F}$. Then $T : V \rightarrow U$ is called a linear transformation of V into U if the following holds for any $\mathbf{u}, \mathbf{v} \in V$ and any scalar $\alpha \in \mathbb{F}$:

- (i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
- (ii) $\mathbf{u} \in V$ it holds that $T(\alpha\mathbf{v}) = \alpha T(\mathbf{v})$.

Example 3. Consider the function $T : V \rightarrow U$ defined by $T(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in V$. Verify that this a linear transformation. This is called the zero transformation.

Example 4. Consider the function $T : V \rightarrow V$ defined by $T(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in V$. Verify that this a linear transformation. This is called the identity transformation.

Example 5. Let $A \in M_{m \times n}(\mathbb{F})$ be given. Consider the function $L_A : \mathbb{F}^n \rightarrow \mathbb{F}^m$ defined by $L_A(\mathbf{v}) = A\mathbf{v}$ for each column vector $\mathbf{v} \in \mathbb{F}^n$. Verify that this a linear transformation. This is called a left multiplication transformation.

Example 6. Is the function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ a linear transformation of $\mathbb{R}^2$ into $\mathbb{R}^3$?

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 - x_2 \\ x_1 + x_2 \\ -x_1 - 3x_2 \end{bmatrix}.$$

Yes! We must verify that the two conditions in Definition 2 hold. For the first condition, take arbitrary vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$. We compute:

$$T(\mathbf{u} + \mathbf{v}) = T\left(\begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}\right) = \begin{bmatrix} 2(u_1 + v_1) - (u_2 + v_2) \\ (u_1 + v_1) + (u_2 + v_2) \\ -(u_1 + v_1) - 3(u_2 + v_2) \end{bmatrix}.$$

Expanding and simplifying, we find

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}).$$

For the second condition, let $c \in \mathbb{R}$ be an arbitrary scalar. Then:

$$T(c\mathbf{u}) = T\left(\begin{bmatrix} cu_1 \\ cu_2 \end{bmatrix}\right) = \begin{bmatrix} c(2u_1 - u_2) \\ c(u_1 + u_2) \\ c(-u_1 - 3u_2) \end{bmatrix} = cT(\mathbf{u}).$$

Therefore, both conditions of Definition 2 hold, and thus T is a linear transformation.

Example 7. Let $V = M_{n \times n}(\mathbb{R})$ be the vector space of $n \times n$ matrices with entries from $\mathbb{R}$ and let $T : V \rightarrow V$ be the function

$$T(A) = A + A^T.$$

Is T a linear transformation of V into V ?

Yes! Let A and B be matrices in V . Then using the properties of the transpose and regrouping, we obtain:

$$T(A + B) = (A + B) + (A + B)^T = A + B + A^T + B^T = (A + A^T) + (B + B^T) = T(A) + T(B).$$

Similarly, if α is any scalar, then

$$T(\alpha A) = (\alpha A) + (\alpha A)^T = \alpha A + \alpha A^T = \alpha(A + A^T) = \alpha T(A).$$

This proves that T satisfies both conditions and thus T is a linear transformation of V into V .

Example 8. Is the function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ a linear transformation of $\mathbb{R}^2$ into $\mathbb{R}^2$?

$$T(\mathbf{x}) = \begin{bmatrix} x_1^2 \sin(x_2) - \cos(x_1^2 - 1) \\ x_1^2 + x_2^2 + 1 \end{bmatrix}$$

No! Recall that

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

If T were linear, then by Definition 2, the following must hold:

$$T\left(\begin{bmatrix} 3 \\ 0 \end{bmatrix}\right) = T\left(3 \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = 3T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = 3 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}.$$

However,

$$T\left(\begin{bmatrix} 3 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3^2 \sin(0) - \cos(3^2 - 1) \\ 3^2 + 0^2 + 1 \end{bmatrix} = \begin{bmatrix} -\cos(8) \\ 10 \end{bmatrix} \neq \begin{bmatrix} -3 \\ 6 \end{bmatrix}.$$

Example 9. Let $V = M_{n \times n}(\mathbb{R})$ be the vector space of $n \times n$ matrices with entries from $\mathbb{R}$, where $n \geq 2$, and let $T : V \rightarrow \mathbb{R}$ be the function

$$T(A) = \det(A).$$

Is T a linear transformation of V into $\mathbb{R}$?

No! If T is a linear transformation, then according to Definition 2, we must have $T(A + B) = \det(A + B) = \det(A) + \det(B)$ and also $T(\alpha A) = \alpha T(A)$ for any scalar α . Do these properties actually hold though?

For example, we know from the properties of the determinant that $\det(\alpha A) = \alpha^n \det(A)$, and therefore it does not hold that $T(\alpha A) = \alpha T(A)$ unless $\alpha = 1$. Therefore, T is not a linear transformation.

Also, it does not hold in general that $\det(A + B) = \det(A) + \det(B)$; in fact, it rarely holds. For example, if

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 1 \\ 0 & 3 \end{bmatrix},$$

then $\det(A) = 2$, $\det(B) = -3$, and therefore $\det(A) + \det(B) = -1$. On the other hand,

$$A + B = \begin{bmatrix} 1 & 1 \\ 0 & 4 \end{bmatrix}$$

and thus $\det(A + B) = 4$. Thus, $\det(A + B) \neq \det(A) + \det(B)$.

Example 10. Let $V = \mathbb{R}_n[t]$ be the vector space of polynomials in the variable t of degree no more than $n \geq 1$. Consider the function $T : V \rightarrow V$ defined as

$$T(f(t)) = 2f(t) + f'(t).$$

For example, if $f(t) = 3t^6 - t^2 + 5$ then

$$T(f(t)) = 2f(t) + f'(t) = 2(3t^6 - t^2 + 5) + (18t^5 - 2t) = 6t^6 + 18t^5 - 2t^2 - 2t + 10.$$

Is T a linear transformation of V into V ?

Yes! Let $f(t)$ and $g(t)$ be polynomials of degree no more than $n \geq 1$. Then

$$\begin{aligned} T(f(t) + g(t)) &= 2(f(t) + g(t)) + (f(t) + g(t))' = 2f(t) + 2g(t) + f'(t) + g'(t) \\ &= (2f(t) + f'(t)) + (2g(t) + g'(t)) \\ &= T(f(t)) + T(g(t)). \end{aligned}$$

Therefore, $T(f(t) + g(t)) = T(f(t)) + T(g(t))$. Now let α be any scalar. Then

$$T(\alpha f(t)) = 2(\alpha f(t)) + (\alpha f(t))' = 2\alpha f(t) + \alpha f'(t) = \alpha(2f(t) + f'(t)) = \alpha T(f(t)).$$

Therefore, $T(\alpha f(t)) = \alpha T(f(t))$. Thus, T is a linear transformation of V into V .

1.3 Kernel and range of a linear transformation

We now introduce two important subsets associated with a linear transformation.

Definition 11. Let V and U be vector spaces over the scalar field $\mathbb{F}$. Let $T : V \rightarrow U$ be a linear transformation of V into U .

- (i) The kernel of T is the set of vectors $\mathbf{v}$ in the domain V that get mapped to the zero vector, that is, $T(\mathbf{v}) = \mathbf{0}$. We denote the kernel of T by $\ker(T)$:

$$\ker(T) = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}\}.$$

It is also known as the null space of T .

- (ii) The range of T is the set of vectors $\mathbf{b}$ in the codomain U for which there exists at least one $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{b}$. We denote the range of T by $\text{Range}(T)$:

$$\text{Range}(T) = \{\mathbf{b} \in U \mid \text{there exists some } \mathbf{v} \in V \text{ such that } T(\mathbf{v}) = \mathbf{b}\}.$$

It is also known as the image of T .

You may have noticed that the definition of the range of a linear transformation is the usual definition of the range of a function. Not surprisingly, the kernel and range are subspaces of the domain and codomain, respectively.

Theorem 12. Let V and U be vector spaces over the scalar field $\mathbb{F}$. Let $T : V \rightarrow U$ be a linear transformation of V into U . Then $\ker(T)$ is a subspace of V and $\text{Range}(T)$ is a subspace of U .

Proof. Suppose that $\mathbf{v}$ and $\mathbf{u}$ are in $\ker(T)$. Then $T(\mathbf{v}) = \mathbf{0}$ and $T(\mathbf{u}) = \mathbf{0}$. By the linearity of T , it holds that

$$T(\mathbf{v} + \mathbf{u}) = T(\mathbf{v}) + T(\mathbf{u}) = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Therefore, since $T(\mathbf{u} + \mathbf{v}) = \mathbf{0}$, $\mathbf{u} + \mathbf{v}$ is in $\ker(T)$. This shows that $\ker(T)$ is closed under addition.

Now suppose that α is any scalar and $\mathbf{v}$ is in $\ker(T)$. Then $T(\mathbf{v}) = \mathbf{0}$, and thus by linearity of T , it holds that

$$T(\alpha\mathbf{v}) = \alpha T(\mathbf{v}) = \alpha \cdot \mathbf{0} = \mathbf{0}.$$

Therefore, since $T(\alpha\mathbf{v}) = \mathbf{0}$, $\alpha\mathbf{v}$ is in $\ker(T)$, which proves that $\ker(T)$ is closed under scalar multiplication.

Lastly, by linearity of T , it holds that

$$T(\mathbf{0}) = T(\mathbf{v} - \mathbf{v}) = T(\mathbf{v}) - T(\mathbf{v}) = \mathbf{0},$$

that is, $T(\mathbf{0}) = \mathbf{0}$. Therefore, the zero vector $\mathbf{0}$ is in $\ker(T)$. This proves that $\ker(T)$ is a subspace of $\mathbf{v}$. **The proof that $\text{Range}(T)$ is a subspace of U is left as an exercise.** $\square$

Example 13. Let $V = M_{n \times n}(\mathbb{R})$ be the vector space of $n \times n$ matrices with entries in $\mathbb{R}$, and let $T : V \rightarrow V$ be the mapping

$$T(A) = A + A^T.$$

Describe the kernel of T .

A matrix A is in the kernel of T if $T(A) = A + A^T = \mathbf{0}$, that is, if $A^T = -A$. Hence,

$$\ker(T) = \{A \in M_{n \times n} \mid A^T = -A\}.$$

What type of matrix A satisfies $A^T = -A$? For example, consider the case where A is the 2×2 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

and $A^T = -A$. Then

$$\begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} = \begin{bmatrix} -a_{11} & -a_{12} \\ -a_{21} & -a_{22} \end{bmatrix}.$$

Therefore, it must hold that $a_{11} = -a_{11}$, $a_{21} = -a_{12}$, and $a_{22} = -a_{22}$. Thus, necessarily $a_{11} = 0$, $a_{22} = 0$, and a_{12} can be arbitrary. For example, the matrix

$$A = \begin{bmatrix} 0 & 7 \\ -7 & 0 \end{bmatrix}$$

satisfies $A^T = -A$.

Using a similar computation as above, a 3×3 matrix satisfies $A^T = -A$ if A is of the form

$$A = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}.$$

where a, b, c are arbitrary constants. In general, a matrix A that satisfies $A^T = -A$ is called skew-symmetric.

Example 14. Let V be the vector space of differentiable functions on the interval $[a, b]$. That is, f is an element of V if $f : [a, b] \rightarrow \mathbb{R}$ is differentiable. Describe the kernel of the linear mapping $T : V \rightarrow V$ defined as

$$T(f(x)) = f(x) + f'(x).$$

A function f is in the kernel of T if $T(f(x)) = \mathbf{0}$, that is, if

$$f(x) + f'(x) = \mathbf{0}.$$

Equivalently, if $f'(x) = -f(x)$. What functions f do you know that satisfy $f'(x) = -f(x)$? How about $f(x) = e^{-x}$? It is clear that $f'(x) = -e^{-x} = -f(x)$ and thus $f(x) = e^{-x}$ is in $\ker(T)$. How about $g(x) = 2e^{-x}$? We compute that $g'(x) = -2e^{-x} = -g(x)$ and thus g is also in $\ker(T)$. It turns out that the elements of $\ker(T)$ are of the form $f(x) = Ce^{-x}$ for a constant C .

Example 15. Consider the linear transformation $T : \mathbb{R}_3[t] \rightarrow M_{2 \times 2}(\mathbb{R})$ defined by:

$$T(p(t)) = p(A),$$

where $A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$. For a polynomial $p(t) = at^3 + bt^2 + ct + d$, the transformation is computed as:

$$T(p(t)) = aA^3 + bA^2 + cA + dI,$$

where I is the identity matrix. The powers of A are:

$$A^2 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}, \quad A^3 = \begin{bmatrix} -2 & 2 \\ -2 & -2 \end{bmatrix}.$$

Thus,

$$T(p(t)) = a \begin{bmatrix} -2 & 2 \\ -2 & -2 \end{bmatrix} + b \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} + c \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} + d \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Kernel of T

To find $\ker(T)$, solve $T(p(t)) = \mathbf{0}$:

$$T(at^3 + bt^2 + ct + d) = \begin{bmatrix} -2a + c + d & 2a + 2b + c \\ -2a - 2b - c & -2a + c + d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

This gives the equations:

$$-2a + c + d = 0, \quad 2a + 2b + c = 0.$$

Solving these, let a and b be free variables. Then:

$$c = -2a - 2b, \quad d = 4a + 2b.$$

Thus, the kernel is the set of all polynomials of the form:

$$p(t) = a(t^3 - 2t + 4) + b(t^2 - 2t + 2),$$

and a basis for $\ker(T)$ is:

$$\{t^3 - 2t + 4, t^2 - 2t + 2\}.$$

Range of T

To find $\text{Range}(T)$, determine the set of all matrices of the form:

$$\begin{bmatrix} w & x \\ y & z \end{bmatrix} = \begin{bmatrix} -2a + c + d & 2a + 2b + c \\ -2a - 2b - c & -2a + c + d \end{bmatrix}.$$

The consistency conditions for w, x, y, z are:

$$x + y = 0, \quad w = z.$$

This implies that $\text{Range}(T)$ consists of matrices of the form:

$$\begin{bmatrix} z & -y \\ y & z \end{bmatrix}.$$

A basis for $\text{Range}(T)$ is:

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \right\}.$$

1.4 The rank-nullity theorem

Definition 16. Let V and U be vector spaces over the scalar field $\mathbb{F}$. Let $T : V \rightarrow W$ be a linear transformation. The dimension of the kernel of T is called the nullity of T and is denoted by $\text{nullity}(T)$. The dimension of the range of T is called the rank of T and is denoted by $\text{rank}(T)$.

The following theorem is a fundamental result about linear transformations.

Theorem 17. Let V be a finite-dimensional vector space and U be vector space over the scalar field $\mathbb{F}$. Let $T : V \rightarrow W$ be a linear transformation. Then

$$\text{nullity}(T) + \text{rank}(T) = \dim(V).$$

Proof. As with almost every proof that involves dimensions, we make use of bases for various vector spaces involved. What we must do is relate the sizes of bases for $\ker(T)$, $\text{Range}(T)$, and V . Since $\ker(T)$ is a subspace of V , it makes sense to start with these two vector spaces ($\ker(T)$ and V). Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a basis for $\ker(T)$ (the smaller of the two spaces). This is a linearly independent set of vectors in V , so we can extend it to a basis for all of V . Let this extended basis be $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$. In forming these two bases, we have labeled the dimensions of $\ker(T)$ and V . That is, we have said that $\dim(\ker(T)) = k$, and $\dim(V) = k + m$. This means that we must show that the dimension of the range of T is m .

Claim: A basis for the range of T is $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_m)\}$. If we can verify this claim, we will have finished the proof. To show this set is a basis, we must establish that this is a spanning set for the range of T and that this set is linearly independent. We tackle these properties in the order listed.

Spanning set of $\text{Range}(T)$: Let $\mathbf{w}$ be in the range of T . This means that $\mathbf{w} = T(\mathbf{v})$ for some $\mathbf{v}$ in

V . This $\mathbf{v}$ can be written as a combination of basis vectors so

$$\mathbf{v} = c_1 \mathbf{u}_1 + \cdots + c_k \mathbf{u}_k + d_1 \mathbf{v}_1 + \cdots + d_m \mathbf{v}_m.$$

Applying the transformation and making use of linearity,

$$\mathbf{w} = T(\mathbf{v}) = T(c_1 \mathbf{u}_1 + \cdots + c_k \mathbf{u}_k + d_1 \mathbf{v}_1 + \cdots + d_m \mathbf{v}_m) = c_1 T(\mathbf{u}_1) + \cdots + c_k T(\mathbf{u}_k) + d_1 T(\mathbf{v}_1) + \cdots + d_m T(\mathbf{v}_m).$$

Since the $\mathbf{u}$'s are all in the kernel of T , we have $T(\mathbf{u}_j) = \mathbf{0}$ for each j . Consequently,

$$\mathbf{w} = d_1 T(\mathbf{v}_1) + \cdots + d_m T(\mathbf{v}_m),$$

which shows that $\mathbf{w}$ is in the span of $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_m)\}$.

Linear Independence: Suppose that $d_1 T(\mathbf{v}_1) + \cdots + d_m T(\mathbf{v}_m) = \mathbf{0}$ for some scalars $d_1, \dots, d_m$. We must show that all the d 's are forced to be 0. By linearity,

$$d_1 T(\mathbf{v}_1) + \cdots + d_m T(\mathbf{v}_m) = \mathbf{0} \implies T(d_1 \mathbf{v}_1 + \cdots + d_m \mathbf{v}_m) = \mathbf{0},$$

so $d_1 \mathbf{v}_1 + \cdots + d_m \mathbf{v}_m$ is in the kernel of T . Since we have a basis for $\ker(T)$, we have

$$d_1 \mathbf{v}_1 + \cdots + d_m \mathbf{v}_m = c_1 \mathbf{u}_1 + \cdots + c_k \mathbf{u}_k.$$

This looks like a linear dependence relation among the $\mathbf{u}$'s and $\mathbf{v}$'s, but the $\mathbf{u}$'s and $\mathbf{v}$'s are linearly independent. The only possibility, then, is that all the coefficients, all the c 's and all the d 's, are zero. In particular, all the d 's must be zero. This shows that $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_m)\}$ is a linearly independent set, completing the proof that it is a basis. $\square$

1.5 The matrix representation of a linear transformation

Until now, we have studied linear transformations by examining their ranges and kernels. Now, we embark on one of the most useful approaches to the analysis of a linear transformation on a finite-dimensional vector space: the representation of a linear transformation by a matrix.

1.5.1 Coordinates

Let V be a finite-dimensional vector space over $\mathbb{F}$. Recall that a basis of V is a set of vectors $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in V such that

- (i) the set B spans all of V , that is, $V = \text{span}(B)$, and
- (ii) the set B is linearly independent.

Hence, if B is a basis for V , each vector $\mathbf{x} \in V$ can be written as a linear combination of vectors in B :

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n.$$

Moreover, from the definition of linear independence, any vector $\mathbf{x} \in \text{span}(B)$ can be written in only one way as a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$. In other words, for the $\mathbf{x}$ above, there does not exist other scalars $t_1, \dots, t_n$ such that also

$$\mathbf{x} = t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \cdots + t_n\mathbf{v}_n.$$

To see this, suppose that we can write $\mathbf{x}$ in two different ways using B :

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n,$$

$$\mathbf{x} = t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \cdots + t_n\mathbf{v}_n.$$

Then

$$\mathbf{0} = \mathbf{x} - \mathbf{x} = (c_1 - t_1)\mathbf{v}_1 + (c_2 - t_2)\mathbf{v}_2 + \cdots + (c_n - t_n)\mathbf{v}_n.$$

Since $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent, the only linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$ that gives the zero vector $\mathbf{0}$ is the trivial linear combination. Therefore, it must be the case that $c_i - t_i = 0$, or equivalently that $c_i = t_i$ for all $i = 1, 2, \dots, n$. Thus, there is only one way to write $\mathbf{x}$ in terms of the vectors in $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Hence, relative to the basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, the scalars $c_1, c_2, \dots, c_n$ uniquely determine the vector $\mathbf{x}$, and vice versa.

We also need the concept of an ordered basis for a vector space.

Definition 18. Let V be a finite-dimensional vector space over $\mathbb{F}$. An ordered basis for V is a basis for V endowed with a specific order; that is, an ordered basis for V is a finite

sequence of linearly independent vectors in V that generates V .

Example 19. In $\mathbb{R}^2$, we have the ordered basis

$$B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}.$$

A different ordered basis is

$$B' = \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}.$$

Our preceding discussion on the unique representation property of vectors in a given basis leads to the following definition.

Definition 20. Let V be a finite-dimensional vector space over $\mathbb{F}$. Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an ordered basis for V . For $\mathbf{x} \in V$, let $c_1, c_2, \dots, c_n$ be the unique scalars such that

$$\mathbf{x} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n.$$

We define the coordinates of $\mathbf{x}$ relative to the ordered basis B to be the unique scalars $c_1, c_2, \dots, c_n$. The coordinate vector of $\mathbf{x}$ relative to the ordered basis B , denoted by $[\mathbf{x}]_B$, is defined to be

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{F}^n.$$

If it is clear what ordered basis we are working with, we will omit the subscript B and simply write $[\mathbf{x}]$ for the coordinates of $\mathbf{x}$ relative to B .

Example 21. One can verify that

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$$

is an ordered basis for $\mathbb{R}^2$. Find the coordinate vector of $\mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ relative to the ordered basis B .

Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. By definition, the coordinates of $\mathbf{v}$ with respect to B are the scalars c_1, c_2 such that

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

If we put $P = [\mathbf{v}_1 \ \mathbf{v}_2]$, and let $[\mathbf{v}]_B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$, then we need to solve the linear system

$$\mathbf{v} = P[\mathbf{v}]_B.$$

Solving the linear system, one finds that the solution is $[\mathbf{v}]_B = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, and therefore this is the coordinate vector of $\mathbf{v}$ relative to the ordered basis B .

It is clear how the procedure of the previous example can be generalized. Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an ordered basis for $\mathbb{R}^n$ and let $\mathbf{v}$ be any vector in $\mathbb{R}^n$. Put $P = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]$. Then the B -coordinates of $\mathbf{v}$ is the unique column vector $[\mathbf{v}]_B$ solving the linear system

$$P\mathbf{x} = \mathbf{v}$$

that is, $\mathbf{x} = [\mathbf{v}]_B$ is the unique solution to $P\mathbf{x} = \mathbf{v}$. Because $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent, the solution to $P\mathbf{x} = \mathbf{v}$ is

$$[\mathbf{v}]_B = P^{-1}\mathbf{v}.$$

We remark that if an inconsistent row arises when you row reduce $[P \ \mathbf{v}]$, then you have made an error in your row reduction algorithm. In summary, to find coordinates with respect to an ordered basis B in $\mathbb{R}^n$, we need to solve a square linear system.

Example 22. Let

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix},$$

and let $B = \{\mathbf{v}_1, \mathbf{v}_2\}$. One can show that B is linearly independent and therefore an ordered basis for $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$. Determine if $\mathbf{x}$ is in W , and if so, find the coordinate vector of $\mathbf{x}$ relative to B .

By definition, $\mathbf{x}$ is in $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ if we can write $\mathbf{x}$ as a linear combination of $\mathbf{v}_1, \mathbf{v}_2$:

$$\mathbf{x} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$$

Form the associated augmented matrix and row reduce:

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

The system is consistent with solution $c_1 = 2$ and $c_2 = 3$. Therefore, $\mathbf{x}$ is in W , and the B -coordinates of $\mathbf{x}$ are

$$[\mathbf{x}]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Example 23. What are the coordinates of

$$\mathbf{v} = \begin{bmatrix} 3 \\ 11 \\ -7 \end{bmatrix}$$

in the standard ordered basis $E = \{e_1, e_2, e_3\}$?

Clearly,

$$\mathbf{v} = \begin{bmatrix} 3 \\ 11 \\ -7 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 11 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - 7 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore, the coordinate vector of $\mathbf{v}$ relative to $\{e_1, e_2, e_3\}$ is

$$[\mathbf{v}]_E = \begin{bmatrix} 3 \\ 11 \\ -7 \end{bmatrix}$$

Example 24. Let $\mathbb{R}_3[t]$ be the vector space of polynomials of degree at most 3.

- (i) Show that $B = \{1, t, t^2, t^3\}$ is an ordered basis for $P_3[t]$.

(ii) Find the coordinates of $v(t) = 3 - t^2 - 7t^3$ relative to B .

The set $B = \{1, t, t^2, t^3\}$ is a spanning set for $\mathbb{R}_3[t]$. In fact, any polynomial $u(t) = c_0 + c_1t + c_2t^2 + c_3t^3$ is clearly a linear combination of $1, t, t^2, t^3$. Is B linearly independent? Suppose that there exist scalars c_0, c_1, c_2, c_3 such that

$$c_0 + c_1t + c_2t^2 + c_3t^3 = 0.$$

Since the above equality must hold for all values of t , we conclude that $c_0 = c_1 = c_2 = c_3 = 0$. Therefore, B is linearly independent, and consequently an ordered basis for $\mathbb{R}_3[t]$. In the ordered basis B , the coordinates of $v(t) = 3 - t^2 - 7t^3$ are

$$[v(t)]_B = \begin{bmatrix} 3 \\ 0 \\ -1 \\ -7 \end{bmatrix}$$

The ordered basis $B = \{1, t, t^2, t^3\}$ is called the standard ordered basis in $P_3[t]$.

Example 25. Show that

$$B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

is an ordered basis for $M_{2 \times 2}(\mathbb{R})$. Find the coordinate vector of $A = \begin{bmatrix} 3 & 0 \\ -4 & -1 \end{bmatrix}$ relative to B .

Any matrix $M = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}$ can be written as a linear combination of the matrices in B :

$$\begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} = m_{11} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + m_{12} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + m_{21} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + m_{22} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

If

$$c_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

then clearly $c_1 = c_2 = c_3 = c_4 = 0$. Therefore, B is linearly independent, and consequently an

ordered basis for $M_{2 \times 2}(\mathbb{R})$. The coordinate vector of A relative to the ordered basis B is given by

$$[A]_B = \begin{bmatrix} 3 \\ 0 \\ -4 \\ -1 \end{bmatrix}.$$

The basis B above is the standard ordered basis of $M_{2 \times 2}$.

1.5.2 Coordinate transformations

Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an ordered basis of $\mathbb{R}^n$ and let $P = [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n] \in M_{n \times n}(\mathbb{R})$. If $\mathbf{x} \in \mathbb{R}^n$ and $[\mathbf{x}]_B$ is the coordinate vector of $\mathbf{x}$ relative to B , then

$$\mathbf{x} = P[\mathbf{x}]_B. \quad (1)$$

Hence, thinking of $P : \mathbb{R}^n \rightarrow \mathbb{R}^n$ as a linear transformation, P maps B -coordinate vectors to coordinate vectors relative to the standard ordered basis of $\mathbb{R}^n$. For this reason, we call P the change-of-coordinates matrix from the ordered basis B to the standard ordered basis in $\mathbb{R}^n$. If we need to emphasise that P is constructed from the ordered basis B we will write P_B instead of just P . Multiplying equation (1) by P^{-1} we obtain

$$P^{-1}\mathbf{x} = [\mathbf{x}]_B.$$

Therefore, P^{-1} maps coordinate vectors in the standard ordered basis to coordinate vectors relative to B .

Example 26. The columns of the matrix P form a basis B for $\mathbb{R}^3$:

$$P = \begin{bmatrix} 1 & 3 & 3 \\ -1 & -4 & -2 \\ 0 & 0 & -1 \end{bmatrix}.$$

- (a) What vector $x \in \mathbb{R}^3$ has B -coordinate vector $[x]_B = (1, 0, -1)$?
- (b) Find the B -coordinate vector of $\mathbf{v} = (2, -1, 0)$.

The matrix P maps B -coordinate vectors to standard coordinate vectors in $\mathbb{R}^3$. Therefore,

$$\mathbf{x} = P[\mathbf{x}]_B = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}.$$

On the other hand, the inverse matrix P^{-1} maps standard coordinates in $\mathbb{R}^3$ to B -coordinates. One can verify that

$$P^{-1} = \begin{bmatrix} 4 & 3 & 6 \\ -1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$$

Therefore, the B -coordinate vector of $\mathbf{v}$ is

$$[\mathbf{v}]_B = P^{-1}\mathbf{v} = \begin{bmatrix} 4 & 3 & 6 \\ -1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ 0 \end{bmatrix}.$$

In general, if V is an n -dimensional vector space over $\mathbb{F}$ and $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an ordered basis for V , we define the coordinate mapping $P : V \rightarrow \mathbb{F}^n$ relative to B as the mapping

$$P(\mathbf{v}) = [\mathbf{v}]_B.$$

Example 27. Let $V = M_{2 \times 2}(\mathbb{R})$ and let $B = \{A_1, A_2, A_3, A_4\}$ be the standard ordered basis for $M_{2 \times 2}(\mathbb{R})$. What is $P : V \rightarrow \mathbb{R}^4$?

Recall,

$$B = \{A_1, A_2, A_3, A_4\} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

Then for any $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, we have

$$P\left(\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\right) = \begin{bmatrix} a_{11} \\ a_{12} \\ a_{21} \\ a_{22} \end{bmatrix}.$$

1.5.3 The matrix of a linear transformation

Let V and W be finite-dimensional vector spaces over $\mathbb{F}$ and let $T : V \rightarrow W$ be a linear transformation. Then by definition, $T(\mathbf{v} + \mathbf{u}) = T(\mathbf{v}) + T(\mathbf{u})$ and $T(\alpha\mathbf{v}) = \alpha T(\mathbf{v})$ for every $\mathbf{v}, \mathbf{u} \in V$ and $\alpha \in \mathbb{F}$. Let $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an ordered basis of V and let $\gamma = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ be an ordered basis of W . Then for any $\mathbf{v} \in V$, there exist scalars $c_1, c_2, \dots, c_n$ such that

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n$$

and thus $[\mathbf{v}]_\beta = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$ is the coordinate vector of $\mathbf{v}$ relative to the ordered basis β . By linearity of the mapping T , we have

$$T(\mathbf{v}) = T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \dots + c_nT(\mathbf{v}_n).$$

Now, each vector $T(\mathbf{v}_j)$ is in W , and therefore, because γ is a basis of W , there are scalars $a_{1,j}, a_{2,j}, \dots, a_{m,j}$ such that

$$T(\mathbf{v}_j) = a_{1,j}\mathbf{w}_1 + a_{2,j}\mathbf{w}_2 + \dots + a_{m,j}\mathbf{w}_m.$$

In other words,

$$[T(\mathbf{v}_j)]_\gamma = \begin{bmatrix} a_{1,j} \\ a_{2,j} \\ \vdots \\ a_{m,j} \end{bmatrix}.$$

Substituting $T(\mathbf{v}_j) = a_{1,j}\mathbf{w}_1 + a_{2,j}\mathbf{w}_2 + \dots + a_{m,j}\mathbf{w}_m$ for each $j = 1, 2, \dots, n$ into

$$T(\mathbf{v}) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \dots + c_nT(\mathbf{v}_n)$$

and then simplifying, we get

$$T(\mathbf{v}) = \sum_{i=1}^m \sum_{j=1}^n c_j a_{i,j} \mathbf{w}_i.$$

Therefore,

$$[T(\mathbf{v})]_{\gamma} = A[\mathbf{v}]_{\mathbf{B}}$$

where A is the $m \times n$ matrix with entries in $\mathbb{F}$ given by

$$A = \begin{pmatrix} [T(\mathbf{v}_1)] & [T(\mathbf{v}_2)] & \dots & [T(\mathbf{v}_n)] \end{pmatrix}.$$

The matrix A is the matrix representation of the linear transformation T in the ordered bases β and γ .

Example 28. Consider the vector space $V = \mathbb{R}_2[t]$ of polynomials of degree no more than two and let $T : V \rightarrow V$ be defined by

$$T(\mathbf{v}(t)) = 4\mathbf{v}'(t) - 2\mathbf{v}(t).$$

It is straightforward to verify that T is a linear mapping. Let

$$B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{t - 1, 3 + 2t, t^2 + 1\}.$$

- (i) Verify that B is a basis of V .
- (ii) Find the coordinates of $\mathbf{v}(t) = -t^2 + 3t + 1$ in the basis B .
- (iii) Find the matrix representation of T in the basis B .

(i) Suppose that there are scalars c_1, c_2, c_3 such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$$

Then expanding and collecting like terms, we obtain

$$c_3t^2 + (c_1 + 2c_2)t + (-c_1 + 3c_2 + c_3) = 0$$

Since the above holds for all $t \in \mathbb{R}$, we must have

$$c_3 = 0, \quad c_1 + 2c_2 = 0, \quad -c_1 + 3c_2 + c_3 = 0$$

Solving for c_1, c_2, c_3 , we obtain $c_1 = 0, c_2 = 0, c_3 = 0$. Hence, the only linear combination of the vectors in B that produces the zero vector is the trivial linear combination. This proves by

definition that B is linearly independent. Since we already know that $\dim(V) = 3$ and B contains 3 vectors, then B is a basis for V .

(ii) The coordinates of $\mathbf{v}(t) = -t^2 + 3t + 1$ are the unique scalars (c_1, c_2, c_3) such that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{v}$$

In this case, the linear system is

$$c_3 = -1, \quad c_1 + 2c_2 = 3, \quad -c_1 + 3c_2 + c_3 = 1$$

and solving yields $c_1 = 1$, $c_2 = 1$, and $c_3 = -1$. Hence, the coordinate vector of $\mathbf{v}$ relative to the ordered basis B is given by

$$[\mathbf{v}]_B = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

(iii) The matrix representation A of T in the ordered basis B is

$$A = \begin{bmatrix} [T(\mathbf{v}_1)]_B & [T(\mathbf{v}_2)]_B & [T(\mathbf{v}_3)]_B \end{bmatrix}.$$

Now we compute directly that

$$T(\mathbf{v}_1) = -2t + 6, \quad T(\mathbf{v}_2) = -4t + 2, \quad T(\mathbf{v}_3) = -2t^2 + 8t - 2$$

and then one computes that

$$[T(\mathbf{v}_1)]_B = \begin{bmatrix} -\frac{18}{5} \\ \frac{4}{5} \\ 0 \end{bmatrix}, \quad [T(\mathbf{v}_2)]_B = \begin{bmatrix} -\frac{6}{5} \\ -\frac{2}{5} \\ 0 \end{bmatrix}, \quad [T(\mathbf{v}_3)]_B = \begin{bmatrix} \frac{24}{5} \\ \frac{8}{5} \\ -2 \end{bmatrix}.$$

And therefore,

$$A = \begin{bmatrix} -\frac{18}{5} & -\frac{6}{5} & \frac{24}{5} \\ \frac{4}{5} & -\frac{2}{5} & \frac{8}{5} \\ 0 & 0 & -2 \end{bmatrix}.$$